MTH 201 Power Series & Taylor's Theorem

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the function $f(x)=\dfrac{1}{1+x}$.

a) Find a power series for $f(x)$ and determine its radius of convergence.

Note that $\dfrac{1}{1+x}=\dfrac{1}{1-(-x)}$. This resembles the sum of a geometric series with $a=1$ and $r=-x$ provided that $|r|<1 \rightarrow |-x|<1 \rightarrow |x|<1 \rightarrow -1<x<1$.

Therefore, $\dfrac{1}{1+x}=\sum_{n=0}^{\infty}(-x)^n=\sum_{n=0}^{\infty}(-1)^n x^n$. This power series has radius of convergence $R=1$.

b) Use your answer from part (a) to determine what function the power series $\sum_{n=1}^{\infty} n(-1)^n x^{n-1}$ models and with what radius of convergence.

We note that this series is just the derivative of the series from part (a). Therefore, the function it models is $f'(x)=\dfrac{-1}{(1+x)^2}$ with $R=1$. Therefore, $\sum_{n=1}^{\infty} n(-1)^n x^{n-1}=\dfrac{-1}{(1+x)^2}$ with $R=1$.

This implies an IOC of at least $(-1,1)$ (possibly including the endpoints).

c) Determine the value of $\sum_{n=1}^{\infty} n(-1)^n \left(\dfrac{3}{4}\right)^{n-1}$. Can you determine the value of $\sum_{n=1}^{\infty} n(-1)^n (2)^{n-1}$? Why or why not?

Since $x=\dfrac{3}{4}$ is within the IOC for the series $\sum_{n=1}^{\infty} n(-1)^n x^{n-1}$ we can say that

$$\sum_{n=1}^{\infty} n(-1)^n x^{n-1}=\frac{-1}{(1+x)^2} \quad \rightarrow \quad \sum_{n=1}^{\infty} n(-1)^n \left(\frac{3}{4}\right)^{n-1}=\frac{-1}{\left(1+\frac{3}{4}\right)^2}=\frac{-1}{\left(\frac{7}{4}\right)^2}=-\frac{16}{49}$$

d) Find a power series centered at $x=8$ for $f(x)=\dfrac{1}{1+x}$ and determine its radius of convergence.

Note that $\dfrac{1}{1+x}=\dfrac{1}{9+(x-8)}=\dfrac{\frac{1}{9}}{1+\frac{(x-8)}{9}}$. This resembles the sum of a geometric series with $a=\dfrac{1}{9}$ and $r=-\dfrac{x-8}{9}$ provided that $|r|<1 \to \left|-\dfrac{x-8}{9}\right|<1 \to |x-8|<9 \to -1<x<17$.

Therefore, $\dfrac{1}{1+x}=\sum_{n=0}^{\infty}\dfrac{1}{9}\left(-\dfrac{(x-8)}{9}\right)^n=\sum_{n=0}^{\infty}(-1)^n\dfrac{(x-8)^n}{9^{n+1}}$. This power series has radius of convergence $R=9$.

e) Which power series (the one from part (a) or part (d)) can be used to create a series that is equal to $f(10)$?

From part (a) we know that $f(x)=\dfrac{1}{1+x}=\sum_{n=0}^{\infty}(-1)^n x^n$ with IOC $(-1,1)$.

From part (d) we know that $f(x)=\dfrac{1}{1+x}=\sum_{n=0}^{\infty}(-1)^n\dfrac{(x-8)^n}{9^{n+1}}$ with a minimum IOC $(-1,17)$.

Therefore, we can use part (d) to establish that $f(10)=\dfrac{1}{11}=\sum_{n=0}^{\infty}(-1)^n\dfrac{(10-8)^n}{9^{n+1}}$.

2. Consider the function $f(x)=\sum_{n=0}^{\infty}\dfrac{x^{n+1}}{n+1}$.

a) Determine what function $f(x)$ resembles and on what interval it resembles this function.

We know from class that $\dfrac{1}{1-x}=\sum_{n=0}^{\infty}x^n$ on the interval $(-1,1)$.

Therefore, $\displaystyle\int\frac{1}{1-x}dx=\int\sum_{n=0}^{\infty}x^n dx \quad\to\quad -\ln|1-x|=C+\sum_{n=0}^{\infty}\frac{x^{n+1}}{n+1}$.

If $x=0$ (which is in the IOC) then $-\ln(1)=C+\sum_{n=0}^{\infty}\dfrac{0}{n+1} \quad\to\quad C=0$.

Therefore $-\ln|1-x|=\sum_{n=0}^{\infty}\dfrac{x^{n+1}}{n+1}$ for all x-values on $(-1,1)$.

So, $f(x)=\sum_{n=0}^{\infty}\dfrac{x^{n+1}}{n+1}$ resembles the function $y=-\ln|1-x|$ on the interval $(-1,1)$.

b) Use your answer from part (a) to help determine the exact value of $\sum_{n=0}^{\infty}\frac{(-1)^{n+1}1/5}{5^n(n+1)}$.

We see that this is the series from part (a) being evaluated at $x=-\frac{1}{5}$,

$$\sum_{n=0}^{\infty}\frac{(-1)^{n+1}1/5}{5^n(n+1)}=\sum_{n=0}^{\infty}\frac{(-1)^{n+1}}{5^{n+1}(n+1)}=\sum_{n=0}^{\infty}\frac{(-1/5)^{n+1}}{(n+1)}$$

Since $x=-\frac{1}{5}$ is in the IOC then we know that $\sum_{n=0}^{\infty}\frac{(-1/5)^{n+1}}{(n+1)}=-\ln\left|1-\left(-\frac{1}{5}\right)\right|=-\ln\left(\frac{6}{5}\right)$

3. Consider the function $f(x)=\sqrt{x}$.

a) Approximate $f(x)$ by a Taylor polynomial with degree 2 centered at $x=4$.

We want $p(x)=f(4)+\frac{f'(4)}{1!}(x-4)+\frac{f''(4)}{2!}(x-4)^2$.

Note that $f'(x)=\frac{1}{2\sqrt{x}}\ \rightarrow\ f'(4)=\frac{1}{4}$ and $f''(x)=-\frac{1}{4x^{3/2}}\ \rightarrow\ f''(4)=-\frac{1}{4(4)^{3/2}}=-\frac{1}{32}$.

Therefore, $p(x)=2+\frac{1}{4}(x-4)-\frac{1}{64}(x-4)^2$

b) Use the polynomial from part (a) to estimate the value of $\sqrt{4.1}$.

$$\sqrt{4.1}=f(4.1)\approx p(4.1)=2+\frac{1}{4}(4.1-4)-\frac{1}{64}(4.1-4)^2=2+\frac{1}{4}(0.1)-\frac{1}{64}(0.1)^2$$

$$=2+\frac{1}{40}-\frac{1}{6400}=\frac{12959}{6400}$$

OR

$$=2+0.025-0.00015625=2.02484375$$

c) Use Taylor's Theorem to estimate the accuracy of the approximation you made in part (b).

By Taylor's Theorem we know that $|R_2(x)|\le\frac{M}{(2+1)!}|x-4|^{2+1}$.

We need $|f'''(x)|\le M$ for all x values within $[4,4.1]$. Note that $f'''(x)=\frac{3}{8x^{5/2}}$ and on the interval $[4,4.1]$ it must be the case that $\frac{3}{8x^{5/2}}\le\frac{3}{8(4)^{5/2}}=\frac{3}{256}$. Therefore, we let $M=\frac{3}{256}$.

$$|R_2(4.1)|\le\frac{\left(\frac{3}{256}\right)}{(3)!}|4.1-4|^3\le\frac{1}{512}|0.1|^3=\frac{1}{512}\left(\frac{1}{10}\right)^3=\frac{1}{512\cdot100}=\frac{1}{51200}=0.00001953125$$

d) Use Taylor's Theorem to estimate the accuracy of the approximation when x is within the interval $[4,4.2]$.

By Taylor's Theorem we know that $|R_2(x)| \le \frac{M}{(2+1)!}|x-4|^{2+1}$.

We need $|f'''(x)| \le M$ for all x values within $[4,4.2]$. Note that $f'''(x) = \frac{3}{8x^{5/2}}$ and on the interval $[4,4.2]$ it must be the case that $\frac{3}{8x^{5/2}} \le \frac{3}{8(4)^{5/2}} = \frac{3}{256}$. Therefore, we let $M = \frac{3}{256}$.

Also, since $4 \le x \le 4.2 \rightarrow 0 \le x-4 \le 0.2$.

Therefore, for x-values in $[4,4.2]$ we have that

$$|R_2(x)| \le \frac{\left(\frac{3}{256}\right)}{(3)!}|x-4|^3 \le \frac{1}{512}|0.2|^3 = \frac{1}{512}\left(\frac{1}{5}\right)^3 = \frac{1}{512\cdot 125} = \frac{1}{64000} = 0.000015625$$

4. Suppose we construct a degree 1 Taylor polynomial centered at $x=3$ for the function $f(x)=\sqrt{1+x}$. What is the maximum possible error of this polynomial when estimating the value of $\sqrt{5}$?

By Taylor's Theorem we know that $|R_1(4)| \le \frac{M}{(1+1)!}|4-3|^{1+1} = \frac{M}{2!}$ where $\left|f^{(1+1)}(x)\right| \le M$.

We consider using Taylor's Theorem on the interval $[3,4]$ and note that on this interval,

$$|f''(x)| = \left|-\frac{1}{4(1+x)^{3/2}}\right| = \frac{1}{4(1+x)^{3/2}} \le \frac{1}{4(1+3)^{3/2}} = \frac{1}{32}$$

Therefore we let $M = 32$.

So, by Taylor's Theorem, the maximum possible error when using this polynomial to estimate $\sqrt{5}$ will be $|R_1(4)| \le \frac{M}{(1+1)!}|4-3|^{1+1} = \frac{M}{2!} = \frac{(1/32)}{2!} = \frac{1}{64}$

5. Suppose you have a mystery function $f(x)$, but you know the following

$$f(1)=2, \quad f'(1)=0, \quad f''(1)=5, \quad f^{(3)}(1)=1, \quad f^{(4)}(x)=\frac{1}{\sqrt{x}5^x}$$

Estimate $f(2)$ to at least $\frac{1}{100}$ accuracy.

Since we have derivatives of *f* evaluated at $x=1$ we can try to approximate $f(x)$ using a power series centered at $x=1$. However, we don't know how many terms of this power series we will need to estimate $f(2)$ to within at least $\frac{1}{100}$.

Since we only have information up through the 4th derivative for *f*, we will try using a degree 3 polynomial to complete the estimation and use Taylor's Theorem to ensure that this will produce the desired level of accuracy.

By Taylor's Theorem we know that $\left|R_3(2)\right| \le \frac{M}{(3+1)!}|2-1|^{3+1}$ where $\left|f^{(3+1)}(x)\right| \le M$.

We consider using Taylor's Theorem on the interval $[1,2]$ and note that on this interval,

$\left|f^{(4)}(x)\right| = \left|\frac{1}{\sqrt{x}5^x}\right| = \frac{1}{\sqrt{x}5^x} \le \frac{1}{5\sqrt{1}} = \frac{1}{5}$. So we will let $M=\frac{1}{5}$.

Therefore we have that $\left|R_3(2)\right| \le \frac{(1/5)}{4!} = \frac{1}{120}$. This means that the error from using a degree 3 polynomial to estimate $f(2)$ will be less than (or equal to) $\frac{1}{120}$ which ensure what we are at least accurate to a level of $\frac{1}{100}$.

So, the degree 3 polynomials that models $f(x)$ is

$$p(x)=2+\frac{0}{1!}(x-1)+\frac{5}{2!}(x-1)^2+\frac{1}{3!}(x-1)^3 = 2+\frac{5}{2}(x-1)^2+\frac{1}{6}(x-1)^3$$

Therefore we can say that $f(2) \approx 2+\frac{5}{2}+\frac{1}{6} = \frac{14}{3}$

6. Suppose we construct a Taylor series centered at $x = \pi$ for the function $f(x) = \sin(x)$ and we want to use this Taylor series to estimate the value of $\sin(6)$ to 4 decimal places of accuracy. How many terms of the Taylor series would we need to use to accomplish level of precision in our estimation?

   By Taylor's Theorem we know that $\left|R_n(3)\right| \le \frac{M}{(n+1)!}\left|3-\pi\right|^{n+1}$ where $\left|f^{(n+1)}(x)\right| \le M$.

   We consider using Taylor's Theorem on the interval $[\pi, 6]$ and note that on this interval (or any interval) $\left|f^{(n+1)}(x)\right| \le 1$ since all derivatives of $f(x)$ are $\pm\sin(x)$ *or* $\pm\cos(x)$ and these functions are all bounded above by 1. So we let $M = 1$.

   Therefore we have $\left|R_n(3)\right| \le \frac{1}{(n+1)!}\left|3-\pi\right|^{n+1}$ and we need $\left|R_n(3)\right| \le \frac{1}{(n+1)!}\left|3-\pi\right|^{n+1} < 0.00001$.

   Since there are several occurrences of *n* here it will be difficult to solve this inequality, so we instead note that $\frac{1}{(n+1)!}\left|3-\pi\right|^{n+1} < \frac{1}{(n+1)!}$ since $\left|3-\pi\right|^{n+1} < 1$.

   So if we can find a value of *n* for which $\frac{1}{(n+1)!} < 0.00001$ then we will have

   $\left|R_n(3)\right| \le \frac{1}{(n+1)!}\left|3-\pi\right|^{n+1} < \frac{1}{(n+1)!} < 0.00001$ as desired.

   We see that $\frac{1}{(n+1)!} < 0.00001 \quad \to \quad (n+1)! > 100{,}000 \quad \to \quad n \ge 8$.

   Thus, if we use at least 8 terms of the Taylor series we can estimate the value of $\sin(6)$ accurately to 4 decimal places.